

MAHIR AFSER PAVEL

AI Researcher & Graduate Assistant Candidate

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Profile Summary

An AI researcher with a strong research interest in **medical image analysis** and **70+ citations** in **5 Q1** and solo Q2 journal publications, 4 of which are **SCIE** and **2 ESCI**, holding an **h-index** and **i10-index** of **2**.

Research Interests

- Vision-Language Models (VLMs) & Large Language Models (LLMs)
- Deep Learning, Explainable Artificial Intelligence (XAI), & Edge AI
- Medical Image Analysis & Computational Clinical Triaging

Education

Degree	Major	Institution	CGPA	Dates Attended
Bachelor of Science	Computer Science and Engineering	North South University	3.71/4.00 (Almost 90% marks)	2019 – 2024

B.Sc. Thesis: *"Non-small cell lung cancer detection through knowledge distillation approach with teaching assistant"*

- **Supervisor:** Dr. Riasat Khan (Associate Professor, Department of ECE).
- **Summary:** Implemented a three-stage knowledge distillation architecture for lung cancer classification on the *NSCLC-Radiomics dataset*, which achieved 94.53% accuracy and was later deployed on a user-friendly web application.

Scholarly Publications

Peer-Reviewed Journal Publications

- **Multi-stage knowledge distillation with layer fusion-based deep learning approach for skin cancer classification.**
✓ **Authors:** Mahir Afser Pavel, Ramisa Asad, Goh Kah Ong Michael, Md Ikramuzzaman, Murad Mustakim, Riasat Khan.
✓ **Venue:** Scientific Reports (Nature Portfolio), 2025. DOI: 10.1038/s41598-025-23403-2
- **Non-small cell lung cancer detection through knowledge distillation approach with teaching assistant**
✓ **Authors:** Mahir Afser Pavel, Rafiul Islam, Shoyeb Bin Babor, Riaz Mehadi, Riasat Khan
✓ **Venue:** PLOS ONE, 2024. DOI: 10.1371/journal.pone.0306441
- **Error-Tolerant Multimodal Vision-Language Models for Endodontic Triaging: A Cross-Sectional Study.**
✓ **Authors:** Mahir Afser Pavel (Co-First Author), Md Fahim Shahoriar Titu, Afifa Zain Apurba, Saif Ahmed, Shafin Rahman, James Dudley, Taseef Hasan Farook
✓ **Venue:** International Journal of Dentistry (Wiley), 2026. DOI: 10.1155/ijod/4148741.
- **Fusion of Image Filtering and Knowledge-Distilled YOLO Models for Root Canal Failure Diagnosis**
✓ **Authors:** Afifa Zain Apurba, Md Fahim Shahoriar Titu, Mahir Afser Pavel, Intisar Tahmid Naheen, Riasat Khan
✓ **Venue:** IEEE Access, 2025. DOI: 10.1109/ACCESS.2025.3560998.
- **Real-time fire detection: Integrating lightweight deep learning models on drones with edge computing**
✓ **Authors:** Md Fahim Shahoriar Titu, Mahir Afser Pavel, Goh Kah Ong Michael, Hisham Babar, Umama Aman, Riasat Khan.
✓ **Venue:** Drones (MDPI), 2024. DOI: 10.3390/drones8090483
- **Hybrid ViT-RetinaNet with Explainable Ensemble Learning for Fine-Grained Vehicle Damage Classification**
✓ **Authors:** Ananya Saha, Mahir Afser Pavel, Md Fahim Shahoriar Titu, Afifa Zain Apurba, Riasat Khan.
✓ **Venue:** Vehicles (MDPI), 2025. DOI: 10.3390/vehicles7030089.

Ongoing Research Publications

- **A Federated Multi-Component Framework for Brain Tumor Diagnosis with Generative Augmentation and Explainable Representation Learning**
 - **Authors:** Mahir Afser Pavel, Nafiz Fahad, Md Tanzib Hosain, Md. Kishor Morol, Tze Hui Liew
 - **Status:** Under Review at Array (array), 2026
- **CLKD-MED: A novel cross-lingual knowledge distillation framework for multilingual clinical outcome prediction**

- Authors: Mahir Afser Pavel, Rafiul Islam, Mohammad Junayed Hasan, and M. R. C. Mahdy
 - Status: Under Review at Information Processing & Management (IPM), 2026
- SLGRA-Net: Structure-Aware Latent Graph Reasoning for 3D Brain Tumor Segmentation
 - Authors: Mahir Afser Pavel, Nafiz Fahad, Md Tanzib Hosain, Md. Kishor Morol, Tze Hui Liew
 - Status: Under Review at Biomedical Signal Processing and Control (BSPC), 2026
- Uncertainty-Aware Explainable Deep Learning Framework for Diabetic Retinopathy Grading Using Monte Carlo Dropout and Grad-CAM
 - Under Review at Conference on Neural Information Processing Systems (NeurIPS), 2026
- Training Dynamics Matter: Reproducibility and Stability in CNN Feature Reliance Evaluation
 - Under Review at British Machine Vision Conference (BMVC), 2026

Professional Experience

Research Assistant | ELITE Research Lab LLC (Dhaka, Bangladesh) (Aug 2025 – Present)

- Federated Medical Architectures: Engineered "Fed-GAN-DE-CL," a federated generative adversarial network utilizing dual-encoder contrastive learning, to enable decentralized brain tumor classification while preserving strict patient data privacy.
- Topological Segmentation: Programmed "SLGRA-Net," incorporating structure-aware latent graph reasoning to enhance boundary definitions in 3D brain tumor MRI segmentations, resolving topological boundary ambiguity.
- Explainable AI Integration: Built an uncertainty-aware diabetic retinopathy grading framework, deploying Monte Carlo Dropout to estimate predictive variance and Grad-CAM for diagnostic explanations.

Research Assistant | North South University (Dhaka, Bangladesh) (May 2024 – Jul 2024)

- Edge Computer Vision: Co-developed real-time drone-based fire detection frameworks, compressing deep models using autodistill-based knowledge distillation to deploy YOLOv8n onto Raspberry Pi 5 edge devices. Improved target detection accuracy by 10-20% under dynamic outdoor constraints.

Research Intern | Mahdy Research Academy (Remote / Online) (Jan 2024 – Mar 2025)

- Clinical Translation Frameworks: Developed "CLKD-MED," a cross-lingual knowledge distillation framework, mapping semantic clinical representations across languages to predict patient outcomes from multilingual electronic health records.

Teaching & Research Experience

- Conducted primary research as a key researcher within the ELITE Research Lab LLC and Mahdy Research Academy, driving the design and execution of deep learning, computer vision, and clinical NLP pipelines.
- Served as an active peer reviewer for IEEE Access, providing rigorous manuscript evaluations for submissions focusing on edge-deployed dual-agent systems and training-free zero-shot action recognition frameworks.
- Co-led hands-on academic sessions and supervised laboratory modules for Analog Electronics and Senior Design at North South University, directly mentoring student groups on circuit mapping, hardware-software co-design, and research publication drafts under faculty supervision.

Standardized Examination Scores

IELTS Academic (2026)	Listening: 7.0	Reading: 7.0	Writing: 6.5	Speaking: 6.0
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Academic Projects & Research Focus

- Quantization-Aware Edge AI: Evaluating and optimizing lightweight deep learning models (YOLOv5, YOLOv8) and multimodal vision-language models (BLIP, CLIP, Florence-2, PaliGemma) using Low-Rank Adaptation (LoRA) and quantization-aware training to minimize memory consumption by up to 87.5% for resource-constrained clinical settings.
- Oncology Image Analytics & Distillation: Refining multi-stage teacher-assistant-student distillation pipelines using Vision Transformers (ViT) and ResNet152v2 on clinical radiography datasets (HAM10000, NSCLC CT scans), enhancing diagnostic sensitivity while preserving model parameters.
- Privacy-Preserving Biomedical Machine Learning: Architecting federated generative adversarial networks and contrastive learning methodologies to facilitate secure, collaborative model training on decentralized, multi-institutional clinical databases.

Technical & Research Skills

Category	Specific Tools and Proficiency Level
Programming Languages	Python (Advanced), SQL, C++, Java, Bash Scripting
AI/ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Detectron2
Generative AI & VLMs	LangChain, LLaMA, RAG, Low-Rank Adaptation (LoRA), QAT, BLIP, CLIP, Florence-2, PaliGemma
Edge AI & Hardware	Raspberry Pi 5, DJI F450 custom drone, Pixhawk PX4, Git, GitHub, LaTeX, Overleaf
Databases & Analysis	SQL, MySQL, NumPy, Pandas, Matplotlib, SPSS, Microsoft Excel

Leadership & Extracurricular Activities

- Research Team Coordinator (2023 – 2024): Supervised a team of 5 research assistants across senior design and directed research projects, coordinating data pipelines and draft submissions that culminated in 5 peer-reviewed journal articles under faculty mentorship.
- Manuscript Reviewer (2026): Provided critical analytical reviews of manuscripts detailing edge AI architectures and zero-shot learning frameworks, maintaining high standards for the journal's methodology criteria.
- Academic Peer Mentor (2022 – 2023): Guided junior engineering students in laboratory environments, resolving hardware-software interfacing issues and assisting in scientific writing workshops

Honors & Awards

- Magna Cum Laude (December 2024):** Conferred by North South University for outstanding academic excellence in the B.Sc. in Computer Science and Engineering program, acknowledging placement within the top 8% in the cohort.
- Academic Merit Scholarship:** Recipient of multiple tuition waivers (10%–20%) from North South University based on high semester grade point averages (semester GPA up to 4.00).
- Peer Reviewer Recognition (March 2026):** Recognized for critical scientific reviews of edge-deployed systems and action recognition manuscripts in IEEE Access and Frontiers in Computer Science.

References

Dr. Riasat Khan, <i>Associate Professor, Dept. of ECE,</i> North South University Email: riasat.khan@northsouth.edu	Dr. Shafin Rahman, <i>Associate Professor, Dept. of ECE,</i> North South University Email: shafin.rahman@northsouth.edu	Md. Kishor Morol, <i>Research Scientist & Founder,</i> ELITE Research Lab LLC Email: kishor@aiub.edu
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